View results

Respondent

47 Anonymous

39:19
Time to complete

A. General information about the respondant

1. * Country

India

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Department of Science and Technology

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://dst.gov.in/

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

Department of Science & Technology (DST) acts as the nodal agency in India for strengthening science, technology and innovation, identifying gap areas in S&T sectors, promoting new and emerging areas of S&T making planning and policy. DST also serves in connecting the science and technology sectors with different Government horizontals and verticals, academia, R&D labs/institutions and industries. DST is responsible for formulation and implementation of the public policies related to the S&T Sector. It provides financial support for research and development in the country to researchers cutting across institutions and disciplines through a competitive mode reinforcing the education system, scientific and industrial R&D and the overall Science, Technology and Innovation landscape of the country.

7. * Full name of officially designated contact person(s) who completed this survey

8. * Position of officially designated contact person(s) who completed this survey

Rabindra Kumar Panigrahy

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Prof. Abhay Karandikar, Secretary, DST

11. Brief description of the consultation process established for the preparation of the report

DST has established a network of Centres for Policy Research (CPR) across the academic institutions in the country and one of the CPR at Indian Institute of Sciences, Banglore is focusing on Open Science as one of its themes of policy research. The DST-CPR at IISc is actively engaged in research on open science, focusing on its value, adoption, and the infrastructural requirements for integrating open science practices into the Indian academic ecosystem. The center's research and capacity-building activities in this area allow it to generate valuable insights directly from its ongoing studies

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

DST-Centre for Policy Research, Indian Institute of Science Bangalore

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

https://dstcpriisc.org/

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [x] Public
- [ ] Private
- [ ] Civil Society
- [ ] Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- (●) Yes
- ( ) No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The Recommendations have been shared with the Departments under Ministry of Science and Technology, O/o Principal Scientific Adviser. to the Government of India

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- ○ Yes
- ◉ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- ◉ Yes
- ○ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

• The DST-CPR at IISc. organises events each year during International Open Access Week to promote open access to research. • Research Data Alliance (RDA's 22nd Plenary – Co-located Events): On May 20, 2024, a virtual panel discussion titled “Advancing Genomic Data Governance: An Indian Perspective” addressed the urgent need for effective governance in India’s genomic research. Organized by Open Science South Asia Network, DST-CPR Indian Institute of Science, UNESCO Chair in Community-Based Research & Social Responsibility in Higher Education, and Transdisciplinary Research Cluster on Frugality Studies-JNU New Delhi, for Research Data Alliance’s Virtual Plenary (14-23 May 2024). The panel aimed to foster the values of data governance and data sovereignty. https://www.rd-alliance.org/rdas-22nd-plenary-co-located-events/ • The Asia OA Meeting, titled "The Promise and Practice of Open Science – The Critical Role of Repositories in Providing Access to Research," was held on April 17-18, 2024, in New Delhi, India. This bi-annual event was organized by the Confederation of Open Access Repositories (COAR) in collaboration with the All India Shri Shivaji Memorial Society College of Engineering (AISSMS) and UNESCO. The meeting brought together policymakers, researchers, and open science experts, including representatives from UNESCO, to discuss strategies for advancing open science in India in a sustainable and equitable manner. Key discussions focused on strengthening open-access infrastructure and overcoming challenges in research dissemination. This gathering followed the G20 Chief Science Advisers’ Roundtable, which emphasized the need for immediate and free access to publicly funded research and the development of interoperability standards for national and international repositories. A major outcome of the meeting was an agreement between COAR and INFLIBNET to collaborate on enhancing India’s open science repository network, reinforcing national efforts toward a more open and accessible research ecosystem. https://coar-repositories.org/news-updates/the-promise-and-practice-of-open-science-coar-asia-oa-meeting/ • SVKM’s NMIMS Deemed-to-be-University hosted a significant International Conference on Open Science as part of the ERASMUS Open Asia Project, which aims to strengthen engagement in Open Science among higher education institutions in India and Malaysia. The event took place at NMIMS Mumbai, with a consortium meeting for partner universities on March 19-20, 2024, followed by the International Conference on March 21-22, 2024. The conference brought together experts and institutions from around the world to discuss Open Science practices, policies, and their impact on higher education and research ecosystems. The partner universities involved in the Open Asia ERASMUS Project included: a. India: Thapar Institute of Engineering and Technology b. Finland: University of Tampere c. Netherlands: VU Amsterdam University d. Malaysia: University of Malaya, University of Malaysia Sabah, and University of Malaysia Sarawak e. Slovenia: International School for Social and Business Studies, and the European Policy Development & Research Institute https://spandan.nmims.edu/2024/04/03/nmims-hosts-erasmus-international-conference-on-open-science-sparks-dialogue-on-the-adoption-of-open-science/

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

- ○ Yes
- ◉ No

22. 1.4. cont. If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

- (selected) Yes
- No

24. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

Title: Science Technology and Innovation Policy 2013. Dept of Science and Technology (DST). Link: https://dst.gov.in/sites/default/files/STI%20Policy%202013-English.pdf). The 2013 policy does not include the above-mentioned element. However, there is a separate policy on Open Access (2013), which mandates Green Open Access. Link: https://dst.gov.in/sites/default/files/APPROVED%20OPEN%20ACCESS%20POLICY-DBT&DST(12.12.2014)_1.pdf

25. 2.1 cont. **If yes:**
  - does it include any of the following open science elements:

- [x] open access to scientific knowledge
- [x] open science infrastructure
- [ ] open engagement of societal actors
- [ ] open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- (•) Yes
- ( ) Partly
- ( ) No

27. 2.2 cont. **If yes or partly,** please share a machine-readable PDF file(s) of the policy(ies) using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

• DST-DBT Open Access Policy (2013) (https://dst.gov.in/sites/default/files/APPROVED%20OPEN%20ACCESS%20POLICY-DBT&DST(12.12.2014)_1.pdf) • Scientific Research Infrastructure Sharing Maintenance and Networks (SRIMAN) The SRIMAN Guidelines aim to promote efficient utilization and wider access of Research Infrastructure (RI) to scientists, researchers, and industry professionals across the country by creating a network of relevant stakeholders. 2022. link: https://dst.gov.in/sites/default/files/SRIMAN%20Guidelines%202022%20Book.pdf • Scientific Social Responsibility Guidelines (SSR). 2022. https://dst.gov.in/sites/default/files/SSR%20Guidelines%202022%20Book_0.pdf

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- (•) Yes
- ( ) Under development
- ( ) No

29. 2.3 cont. **If yes or under development,** please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

> One Nation One Subscription Plan. Union Cabinet approved the ONOS scheme on 25th Nov 2024. A total of almost INR 6,000 crore has been allocated for One Nation One Subscription for 3 calendar years, 2025, 2026 and 2027 as a new Central Sector Scheme. ONOS aims to provide country-wide access to international high impact scholarly research articles and journal publications to students, faculty and researchers of all Higher Education Institutions managed by the central government and state governments and Research & Development Institutions of the central government Link: https://pib.gov.in/PressReleasePage.aspx?PRID=2089179#:~:text=The%20policy%20seeks%20to%20cultivate,on%2025th%20Nov%202024.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

- ○ Yes
- ○ Under development
- ◉ No

31. 2.4 cont. **If no,** please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

> No specific response to offer

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

- (selected) Yes
- Under development
- No

33. 2.5 cont. **If yes,** please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors). To view the elements of open science, see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9

> Some institutions maintain repositories where all the research articles and thesis are deposited (example: https://eprints.iisc.ac.in/). However, public access is limited mainly due to journal embargo policies. Some institutions are working on maintaining and sharing research data. There are several citizen science projects running at various institutions (example: https://citsci-india.org/citizen-science-projects/).

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ○ Yes
- ○ Under development
- ◉ No

35. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

No specific response to offer

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

- ○ Yes
- ● No

37. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

- ○ Yes
- ● No

39. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ○ Yes
- ○ Under development
- ◉ No

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

41. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

• India's GERD as percentage of GDP remained at 0.66% and 0.64% during the years 2019–20 and 2020–21, respectively.
https://dst.gov.in/sites/default/files/Updated%20RD%20Statistics%20at%20a%20Glance%202022-23.pdf

42. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

-There were 751.5 million internet users in India in January 2024. India's internet penetration rate stood at 52.4 percent of the total population at the start of 2024. Link:
https://datareportal.com/reports/digital-2024-india#:~:text=There%20were%20751.5%20million%20internet%20users%20in%20India%20in%20January,January%202023%20and%20January%202024

43. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- (selected) Yes
- No

44. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The Education and Research Network (ERNET, https://ernet.in) India is an autonomous scientific society under the administrative control of Ministry of Information Technology, Government of India having one of the largest nationwide terrestrial and satellite network with nine points of presence located at the premier academic and research institutions in major cities of the country. Focus of ERNET India is not limited to just providing connectivity, but to meet the entire needs of the academic and research institutions by providing consultancy, project management, training and other value added services such as web hosting, e-mail services, video conferencing, domain registration, CUG services. The ERNET India is serving more than thousands of institutions in various sectors, namely health, agriculture, higher education, schools and science & technology. (source: https://www.indiascienceandtechnology.gov.in/organisations/ministry-and-departments/ministry-electronics-and-information-technology-meity-govt-india/education-and-research-network-ernet-new-delhi) National Knowledge Network (NKN) is a state-of-the-art Pan-India network and is a revolutionary step towards creating a knowledge society without boundaries. It provides unprecedented benefits to the knowledge community. Using NKN, all vibrant institutions with vision and passion are able to transcend space and time limitations in accessing information and knowledge and derive the associated benefits for themselves and for the society. It facilitates the development of India's information infrastructure, stimulate research, and create next generation applications and services. The target users for the NKN are all institutions engaged in the generation and dissemination of knowledge in various areas, such as research laboratories, universities and other institutions of higher learning, including professional institutions. NKN has already connected 1808 institutions. https://nkn.gov.in/en/about-us-lt-en

45. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- (•) Yes
- ( ) Under development
- ( ) No

46. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- [x] federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- [ ] community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- [ ] platforms for exchanges and co-creation of knowledge between scientists and society
- [ ] community-based monitoring and information systems to complement national, regional and global data and information systems

47. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

(Sharing of Research Infrastructure: The Indian Science, Technology, and Engineering Facilities Map (I-STEM) is an interactive web designed to accelerate scientific progress in India, link: https://www.istem.gov.in). National Supercomputing Mission (NSM) - The Mission envisages empowering academic and R&D institutions spread over the country by installing a vast supercomputing grid comprising of more than 70 High-Performance Computing (HPC) facilities. These supercomputers will also be networked on the National Supercomputing grid over the National Knowledge Network (NKN). The NKN connects academic institutions and R&D labs over a high speed network. Academic and R&D institutions as well as key user departments/ministries would participate by using these facilities and develop applications of national relevance. https://www.indiascienceandtechnology.gov.in/st-visions/national-mission/national-supercomputing-mission nsm#:~:text=These%20supercomputers%20will%20also%20be,over%20a%20high%20speed%20network.

48. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ○ Yes
- ◉ No

49. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

- ☐ Yes
- ☐ No
- ☑ Not sure about

50. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- ○ Yes
- ◉ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

51. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ○ Yes
- ◉ No

52. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

53. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- [ ] Yes
- [x] No

54. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

55. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

- [ ] Yes
- [x] No

56. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

57. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ○ Yes
- ◉ Partly
- ○ No

58. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ○ Yes
- ◉ No

59. 4.5 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

'

60. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- ○ Yes
- ◉ No

61. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

62. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- ◉ Yes
- ○ No

63. 5.2 cont. **If yes**, please provide details and links as relevant.

Scientific Social Responsibility Guidelines.
https://dst.gov.in/sites/default/files/SSR%20Guidelines%202022%20Book_0.pdf

64. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

- ○ Yes
- ◉ No

65. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

66. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

- ○ Yes
- ◉ No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

67. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- ○ YES, at the national level
- ○ YES, at the institutional level
- ○ YES, at both the national and institutional levels
- ◉ No

68. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

69. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

The CPR at Indian Institute of Sciences, Banglore engages in collaborative research projects with international institutions. It regularly organizes and participates in international workshops and discussions related to STI policy and Open Science. These events provide platforms for researchers and policymakers from different countries to interact and share insights. The DST CPR received funding from the US-based organization Code for Science and Society to promote open science and organized the Open Science South Asia Network Conference in 2022, a first-of-its-kind event in South Asia, to cultivate a network committed to Open Science practices

70. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ○ Yes
- ◉ No

71. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

72. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

- ○ Yes
- ◉ No

73. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

No specific response to offer

Part 8. GENERAL CONSIDERATIONS

74. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- ○ Yes
- ◉ No

75. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

Nothing specific